

Data Dignity Project

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Introduction

As personal data collection becomes increasingly pervasive, users are often left without clear visibility into how their information is stored, shared, or exploited. Existing data management and privacy tools tend to prioritize system efficiency or legal compliance over human understanding, resulting in interfaces that are difficult to navigate, cognitively overwhelming, and inaccessible to users with limited attention tolerance or executive function challenges. This project explores how principles of cognitive accessibility, transparency, and user agency can be applied to the design of a mobile application that restores meaningful control over personal data.

The Data Dignity application was designed to help users understand their digital footprint through clear visual hierarchies, card-based layouts, and layered control systems. Grounded in accessibility research and user-centered design, the project focuses on a teen persona, Jordan, whose needs highlight the consequences of cognitive overload in complex digital systems. This paper documents the full design process, including persona development, research synthesis, use case creation, dashboard rationale, data management architecture, accessibility decisions, visual system design, and future considerations.

The modern digital ecosystem is built on continuous data collection. From mobile applications and wearable devices to educational platforms and financial services, users are constantly generating data, often without fully understanding what is collected, where it is stored, or how it is used. While regulatory frameworks such as GDPR and data transparency initiatives aim to address these issues, the interfaces through which users are expected to exercise control frequently remain inaccessible and cognitively demanding.

Many privacy and data management tools rely on dense lists, technical terminology, and deep navigation hierarchies. These patterns disproportionately disadvantage users who struggle with sustained attention, information overload, or unfamiliar technical language. As a result, data control becomes a symbolic feature rather than a practical one.

The Data Dignity project responds to this problem by reframing data management as a human-centered experience. Instead of presenting raw system data, the application translates complex data relationships into visually scannable, action-oriented interfaces. The goal is not merely to comply with privacy standards, but to ensure that users feel capable, informed, and confident when interacting with their data.

User Persona and Workflow

1- Persona

PERSONA Template

TARGET PERSONA IMAGE	PERSONA INFORMATION
	PERSONA NAME :
	Jordan
	OCCUPATION:
	High School Student
	LOCATION
	Indianapolis, IN
	AGE
	15
	QUOTATION
	"If I can't understand it in a few seconds, I usually just stop using it."

GOALS AND MOTIVATIONS
Detail at least 3 important goals and motivations
• Jordan wants visibility into where their data is stored, sold, or data-mined, without needing to understand complex technical details. • Jordan values autonomy and prefers systems that allow quick, confident decisions rather than requiring parental or expert intervention. • Jordan is motivated by systems that feel interactive and rewarding, especially when progress or improvements are visually reinforced.

VALUES AND WORK METHODS
Describe what's important to your persona, e.g., their work ethic, attention to detail, punctuality, adherence to deadlines, and so on.
• Clarity over completeness • Speed over precision • Visual cues over written explanations

CHARACTERISTICS THAT APPLY TO YOUR PRODUCT
Describe anything relevant to how the person might use your product or similar products or how they currently do the work for which your product is intended.
• Needs low cognitive load interfaces • Prefers card-based layouts and toggles • Benefits from color-coded risk indicators • Avoids deep navigation hierarchies • More likely to act when controls are immediately visible

The primary persona for this project is **Jordan**, a teenage user navigating school, social platforms, fitness tools, and digital services simultaneously. Jordan values independence and dislikes systems that feel restrictive or overly instructional. However, Jordan can quickly become overwhelmed by dense information or complex navigation paths.

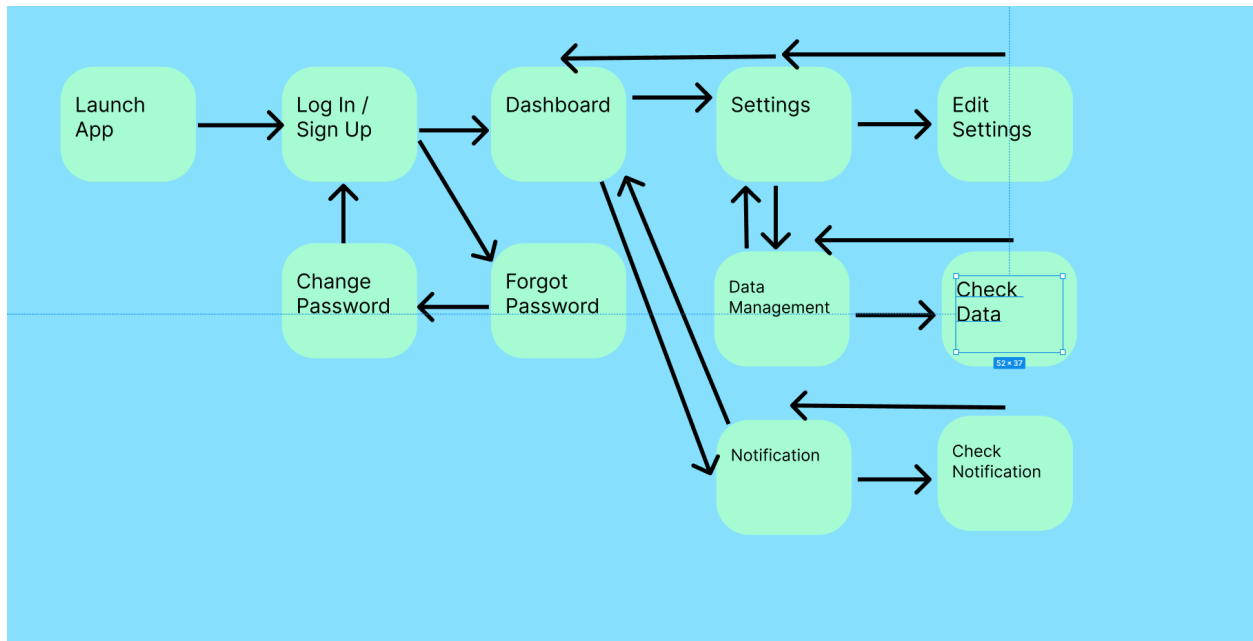
Key characteristics of Jordan include:

- Limited patience for multi-step processes, especially when intermediate steps do not provide clear feedback or value
- A strong preference for visual feedback such as icons, color cues, progress indicators, and status badges over text-heavy explanations
- Difficulty maintaining focus in cluttered or inconsistent interfaces, where competing visual elements increase cognitive load
- A strong desire for autonomy and control, including the ability to act quickly and reverse decisions if needed

Jordan represents a growing demographic of users who are digitally fluent but cognitively burdened by poorly designed systems. Designing for Jordan requires minimizing friction, reducing unnecessary choices, and making actions obvious and reversible.

Research on ADHD-friendly and cognitively accessible design supports this approach, emphasizing the importance of clear task boundaries, predictable layouts, and simplified decision-making structures.

2- Application Workflow



The application workflow illustrated outlines the primary navigation paths a user follows when interacting with the Data Dignity application. The workflow was intentionally designed to support clarity, low cognitive load, and user control, aligning with the needs of the primary persona, Jordan, a teen user who can become overwhelmed by complex or cluttered systems.

The workflow begins when the user launches the app, leading directly into the Log In / Sign Up screen. This entry point provides access to both authentication and account recovery paths, including Forgot Password and Change Password, ensuring that users can recover access without leaving the app or navigating external systems. These recovery paths loop cleanly back into the login flow, reinforcing predictability and reducing frustration during error states.

Once authenticated, users arrive at the Dashboard, which serves as the central hub of the application. From this screen, users can orient themselves and choose their next action without being forced into immediate decisions. The dashboard provides access to three primary areas:

Settings, Data Management, and Notifications. This hub-and-spoke structure minimizes disorientation by ensuring that users can always return to a familiar reference point.

The Settings path allows users to edit global preferences through the Edit Settings screen. This separation between viewing settings and editing them prevents accidental changes and supports Jordan's need for forgiveness and control. Users can easily navigate back to the dashboard from settings, maintaining a consistent mental model of the system.

The Data Management path supports deeper inspection of personal data. From this screen, users can check their data, review how it is stored or used, and then return to the Data Management overview or back to Settings if adjustments are needed. This bidirectional flow reinforces the relationship between global preferences and granular data control without merging them into a single, overwhelming interface.

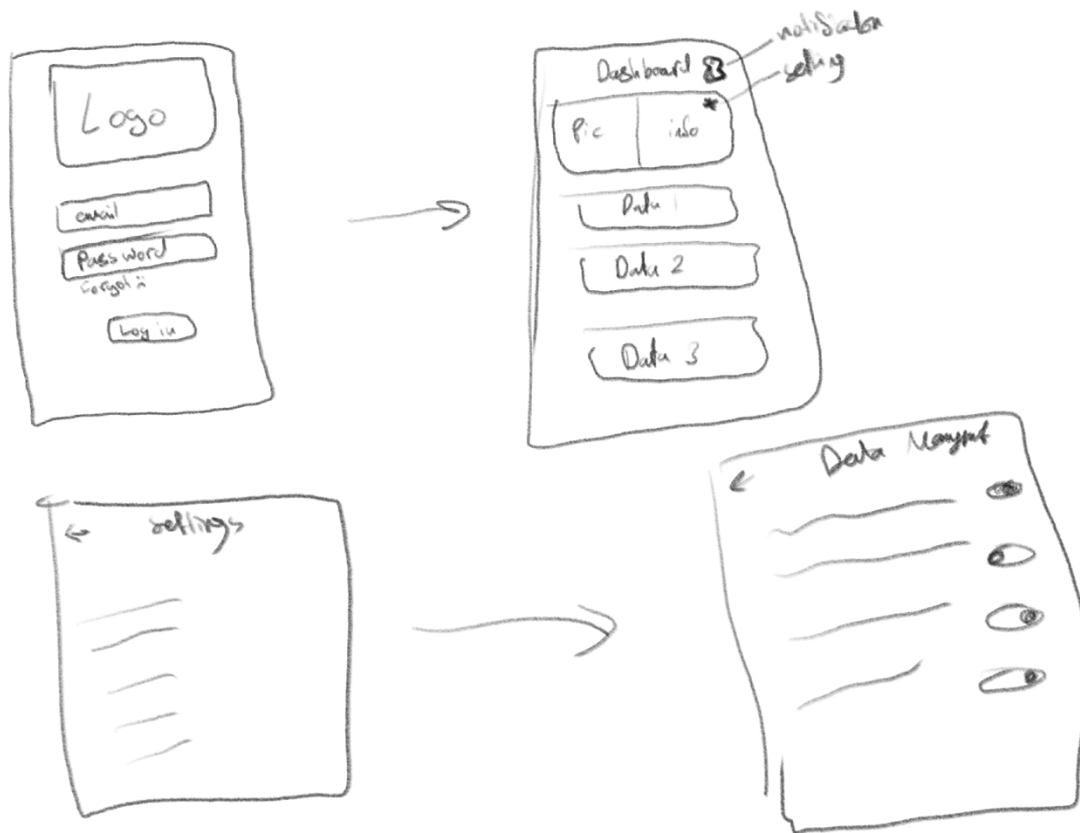
The Notification path allows users to review alerts and system updates. Selecting Check Notification enables users to inspect specific alerts, such as data activity or potential risks, before returning to the notification list or dashboard. This ensures that alerts act as entry points into action rather than dead ends.

Overall, the workflow emphasizes progressive disclosure, allowing complexity to surface only when the user explicitly seeks it. Clear return paths, limited decision points, and consistent navigation patterns reduce cognitive strain and support Jordan's need for autonomy and confidence. By making all major actions reversible and keeping navigation shallow, the application workflow transforms data awareness and control into an approachable, user-centered experience.

Sketches and Wireframes

1- Sketches

Part 1



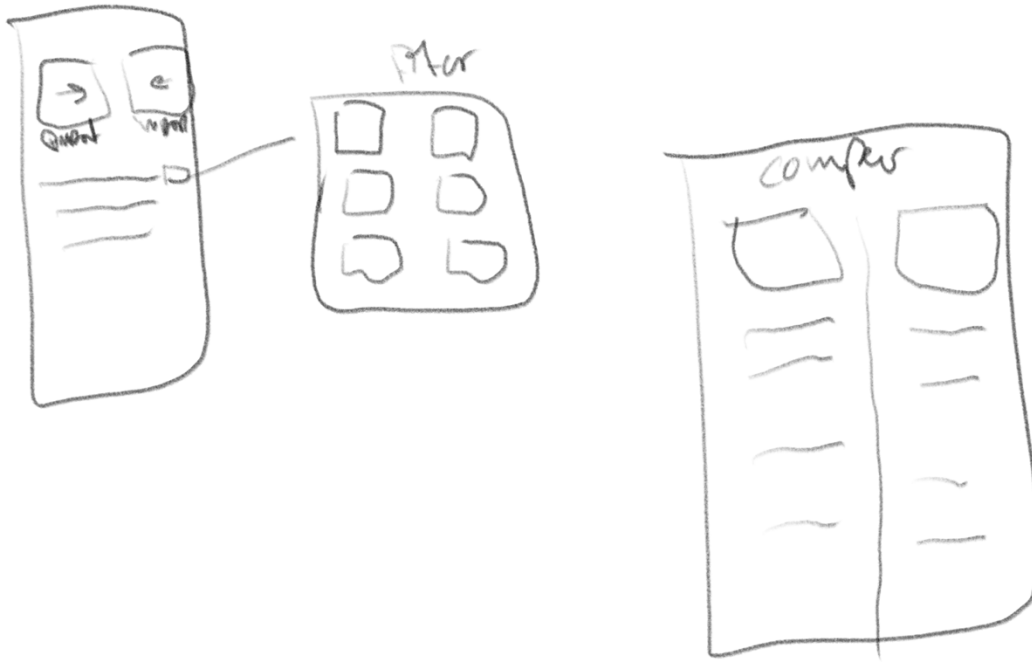
The first set of sketches focuses on establishing the overall structure and navigation logic of the Data Dignity application before committing to detailed layouts or visual styling. These early sketches explore how core sections of the app, such as the dashboard, data categories, and comparison views, relate to one another and how users transition between high-level overviews and more detailed information.

At this stage, the emphasis was on conceptual clarity rather than visual polish. The sketches intentionally use simple shapes and placeholders to represent screens and content blocks. This allowed for rapid exploration of navigation paths and mental models without introducing unnecessary cognitive noise. For example, the dashboard is represented as a compact, centralized screen that branches into category groupings, reinforcing the idea that the dashboard serves as a hub rather than a destination.

The comparison sketch explores how information might be viewed side-by-side to support quick evaluation. This is particularly relevant for a user like Jordan, who benefits from visual organization and struggles with dense lists. By sketching comparison views early, the design process prioritized scalability and reduced reliance on memory when evaluating multiple data sources.

Overall, Part 1 sketches helped validate the information architecture of the app. They ensured that the core navigation felt intuitive, reversible, and shallow, aligning with Jordan's limited patience for deep hierarchies and multi-step navigation.

Part 2



The second set of sketches shifts from high-level structure to screen-level interaction design, focusing on how users move through key moments of the application experience. These sketches include the login screen, dashboard, settings, and data management views, illustrating a clear progression from entry to action.

The login sketch establishes a simple, distraction-free entry point with minimal input fields and a clear call to action. This supports cognitive accessibility by reducing decision-making at the point of entry. From there, the flow leads directly to the dashboard, which is sketched with stacked cards representing different data categories. This reinforces visual hierarchy and allows users to quickly identify areas of interest without reading extensive text.

The settings sketch explores a clean, list-based layout that separates global preferences from data-specific controls. This distinction is intentional and supports the broader design decision to separate Settings from Data Management. The data management sketch then introduces toggle-based controls, allowing users to inspect and modify individual data streams through direct, visible actions rather than hidden menus.

Throughout Part 2, arrows between sketches indicate the intended interaction flow, showing how users move between overview and control states. These transitions were designed to be predictable and forgiving, ensuring that users can always return to a familiar screen such as the dashboard or settings without losing context.

Together, these sketches allowed for early testing of layout, hierarchy, and interaction patterns while keeping cognitive load low. By resolving navigation and interaction questions at the sketching stage, the design process reduced the risk of usability issues later in high-fidelity wireframes and prototypes.

Sketching Summary

The sketching phase played a critical role in shaping the Data Dignity application by allowing design decisions to be explored, tested, and refined at a low level of commitment. Through both high-level conceptual sketches and detailed screen-level layouts, this phase helped surface potential usability issues early while keeping the focus on user needs rather than visual polish.

The Part 1 sketches clarified the overall structure of the application, ensuring that navigation paths were intuitive, shallow, and reversible. By prioritizing clear relationships between the dashboard, data categories, and comparison views, these sketches validated the information architecture and reinforced the dashboard's role as a central hub. This was especially important

for supporting Jordan, whose attention challenges make deep hierarchies and unclear navigation particularly problematic.

The Part 2 sketches shifted attention to interaction flow and layout, exploring how users move from entry points such as login into meaningful actions within settings and data management. These sketches emphasized visual hierarchy, direct controls, and predictable transitions, all of which contribute to reducing cognitive load. By representing controls as toggles and grouping related actions together, the sketches ensured that user control remained visible and accessible rather than hidden behind menus.

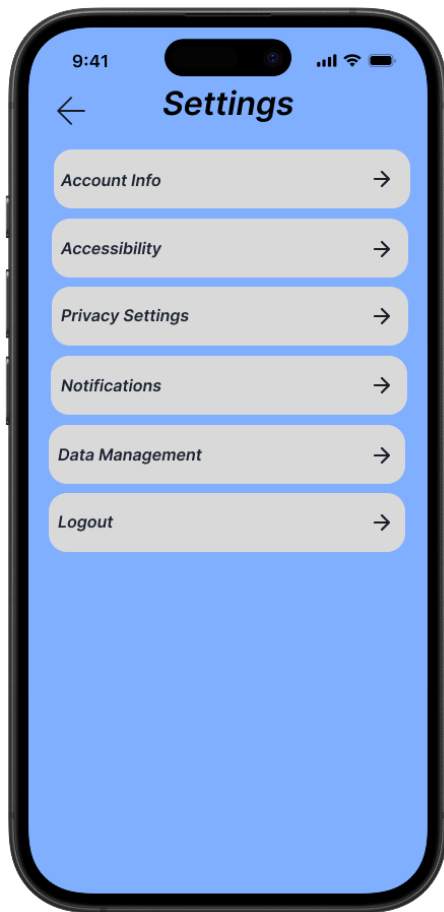
Overall, sketching served as a foundational step that aligned the application's structure and interactions with principles of cognitive accessibility and user autonomy. Resolving navigation, hierarchy, and interaction questions at this stage allowed later wireframes and high-fidelity designs to focus on refinement rather than correction. As a result, the sketches established a clear, user-centered direction for the remainder of the design process.

2- Wireframes



The dashboard screen serves as the primary entry point and orientation hub for the Data Dignity application. It provides Jordan with an immediate, high-level overview of their digital footprint without requiring deep interaction or technical understanding. At the top of the screen, a personalized profile card reinforces a sense of ownership and identity, grounding the experience in the user rather than the system. Below this, data is presented through visually distinct cards that summarize key areas such as overall data distribution, security activity, location sharing, purchase history, and social media data. The use of simple charts and progress bars allows Jordan to quickly scan patterns and risk levels without relying on dense text. Information icons on each

card offer optional context, supporting progressive disclosure and preventing cognitive overload. By combining personalization, clear visual hierarchy, and scannable data summaries, the dashboard enables Jordan to understand their data at a glance and decide where to take action next with confidence.



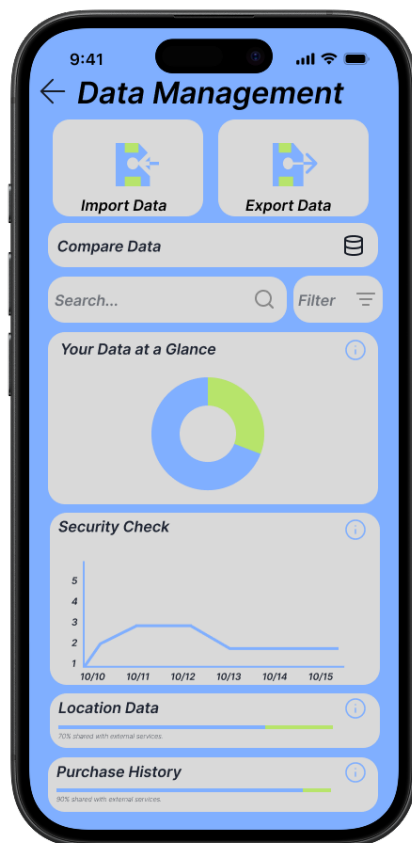
The Settings screen functions as the global control center for the Data Dignity application, allowing users to manage overarching preferences without interfering with specific data streams. The layout uses a simple, vertically stacked list of clearly labeled options, reducing visual clutter and minimizing cognitive load for users like Jordan. Each settings category, such as Account Info, Accessibility, Privacy Settings, Notifications, and Data Management, is presented as a distinct, tappable row, reinforcing predictability and ease of navigation. This structure supports

Jordan's need for clarity by keeping actions shallow and immediately visible, avoiding nested or hidden menus. The inclusion of Accessibility as a first-class option emphasizes the app's commitment to inclusive design, while the separation between Privacy Settings and Data Management reinforces the layered control model used throughout the application. Overall, the Settings screen provides a calm, organized space where users can confidently adjust preferences, knowing that changes are intentional, reversible, and clearly scoped.



The Notification screen provides users with a centralized, chronological view of system messages, alerts, and data-related updates. Notifications are presented as clearly separated cards with concise titles and supporting information, allowing Jordan to quickly scan and prioritize attention without being overwhelmed by dense text. The consistent layout and spacing reduce

visual noise, while the optional “view” icon on each notification enables users to inspect details only when needed, supporting progressive disclosure. This design ensures that alerts act as informative entry points rather than interruptions, allowing Jordan to remain in control of when and how to engage with additional information. By organizing notifications in a predictable, scannable list and avoiding intrusive visual cues, the screen supports cognitive accessibility and helps users stay informed without contributing to notification fatigue.



The Data Management screen is the primary space where users engage directly with their personal data, transitioning from high-level awareness to actionable control. At the top of the screen, prominent actions such as Import Data and Export Data provide immediate access to critical data portability functions, reinforcing transparency and user ownership. The Compare Data option allows users to evaluate multiple data sources side-by-side, reducing reliance on

memory and supporting informed decision-making. Search and filter tools further support efficient navigation by allowing users to narrow focus without introducing additional cognitive burden.

Below these controls, data is presented through a series of visually consistent cards that mirror the structure of the dashboard, including summaries of overall data distribution, security trends, and category-specific sharing levels. This continuity helps maintain a stable mental model as users move between overview and management contexts. Information icons on each card enable optional exploration without forcing interaction, supporting progressive disclosure. By combining direct actions, scannable visualizations, and predictable layout patterns, the Data Management screen empowers Jordan to inspect, compare, and control personal data with confidence while minimizing cognitive load.

Mood board



The mood board was created to establish a clear emotional and visual foundation for the Data Dignity application before finalizing interface details. The selected imagery, color palette, and visual references emphasize calmness, motivation, and reassurance, reflecting the goal of making data awareness feel approachable rather than intimidating. Because the primary persona, Jordan, can be easily overwhelmed by visually dense or aggressive interfaces, the mood board intentionally avoids harsh contrasts and anxiety-inducing imagery often associated with privacy or security tools.

Soft blues and muted accent colors were chosen to convey trust, stability, and clarity, while brighter accent tones are used sparingly to highlight progress, status, and positive reinforcement. Several images focus on progress tracking, daily reminders, and step-by-step completion, reinforcing the idea that managing data is an ongoing, manageable process rather than a one-time, high-stakes task. This aligns with Jordan's preference for visually engaging, gamified systems that provide encouragement and feedback without feeling patronizing.

Human-centered imagery within the mood board reinforces the idea that the application is designed around real users and everyday experiences, not abstract systems. Illustrations and visuals depicting guidance, checkmarks, and gradual progress informed the use of card-based layouts, progress indicators, and confirmation states throughout the interface. Overall, the mood board served as a visual anchor that guided design decisions related to color, tone, and interaction style, ensuring that the final interface supports cognitive accessibility, reduces stress, and reinforces a sense of control and confidence for the user.

Addressing the Project Brief

The Data Dignity project was designed in direct response to the project brief, which emphasized creating an application that helps users understand, manage, and control their personal data in a way that feels transparent, empowering, and accessible. A key requirement of the brief was to support users who may feel overwhelmed by complex systems, while still providing meaningful and actionable control over data visibility and sharing.

To address this, the project centers on the persona Jordan, a teen user with attention and cognitive load challenges. This focus ensured that design decisions consistently prioritized clarity, simplicity, and reduced friction. Rather than presenting dense data tables or technical language, the application translates complex data concepts into visual summaries, card-based layouts, and color-coded indicators that allow users to quickly grasp key information immediately.

The brief also required clear separation between awareness and action. This was addressed through a layered structure consisting of the Dashboard, Notifications, Settings, and Data Management screens. The Dashboard provides high-level awareness, Notifications surface timely and relevant updates, Settings control global preferences, and Data Management enables granular, data-specific actions such as importing, exporting, comparing, or reviewing data streams. This separation prevents cognitive overload while still ensuring full user control.

Accessibility and usability were central to the brief and were addressed through consistent interaction patterns, shallow navigation hierarchies, and progressive disclosure. Users are never forced to engage with more information than necessary, and all actions provide clear feedback and confirmation.

User Testing, Feedback and next steps

User testing was conducted to evaluate how effectively the Data Dignity application supports clarity, control, and cognitive accessibility for its intended users. The primary focus of testing was to observe how users navigated the dashboard, interpreted data visualizations, and completed key tasks within the Settings, Notifications, and Data Management screens. Attention was given to identifying moments of confusion, hesitation, or cognitive overload, especially given the needs of the primary persona, Jordan.

Overall, feedback indicated that users found the application intuitive and visually approachable. Participants responded positively to the card-based layout and consistent visual structure, noting that information felt organized and easy to scan. The dashboard was frequently cited as helpful for providing a quick sense of “what matters right now,” allowing users to identify areas of concern without reading extensive text. Users also appreciated the use of visual indicators, such as charts and progress bars, which reduced the need to interpret raw data.

The separation between Settings and Data Management was largely successful in supporting clear mental models. Users understood Settings as a place for global preferences and Data Management as the space for direct interaction with personal data. However, some users initially expected to find certain data-related controls within Settings, suggesting that clearer labeling or brief explanatory cues could further strengthen this distinction.

Feedback on the Notification screen highlighted its non-intrusive nature. Users valued being able to review alerts at their own pace rather than being interrupted by aggressive prompts. That said, some participants expressed interest in clearer prioritization of notifications, such as visual differentiation between informational messages and higher-risk alerts.

Based on user feedback, several next steps were identified. Future iterations should explore enhanced notification prioritization, additional inline explanations for complex data concepts, and expanded accessibility options that allow users to further customize visual density.

Additional usability testing with a broader range of users, including those with varying cognitive and accessibility needs, would also help validate the design at scale.

In summary, user testing confirmed that the current design direction effectively supports user confidence and reduces cognitive load, while also highlighting opportunities for refinement.

These insights will guide future iterations to further strengthen Data Dignity's goal of making data transparency approachable, actionable, and user centered.

Design elements, interactions, and any specific functionality

1- Design Elements

The interface relies on a card-based design system that organizes information into visually distinct, manageable sections. Cards are used consistently across the dashboard, data management, and notification screens to group related content and reduce visual clutter. This structure supports quick scanning and allows users to focus on one piece of information at a time.

Color is used functionally rather than decoratively. Soft blue backgrounds establish a calm, trustworthy tone, while neutral grays separate content without overwhelming the user. Accent colors such as green and yellow are reserved for status indicators, signaling safe or attention-worthy data states. Icons are simple and recognizable, reinforcing meaning without requiring text-heavy explanations.

Typography is clean and legible, with clear hierarchy between screen titles, section headers, and supporting text. Plain-language labels are used throughout to avoid technical jargon, ensuring that content is accessible to users with varying levels of technical literacy.

2- Interactions

Interactions within the app are designed to be **direct, visible, and forgiving**. Toggle switches are used for binary actions such as enabling or disabling data streams, allowing users to immediately see the current system state. Buttons and tappable cards provide clear affordances, making it obvious what elements are interactive.

Progressive disclosure is a core interaction principle. High-level summaries are presented first, with additional detail revealed only when users choose to engage further. Information icons and “view” actions allow users to explore context on demand without being forced into complex screens.

Feedback is provided consistently after every significant action. Confirmation states, subtle visual changes, and success indicators reassure users that their actions have been completed successfully. Where appropriate, actions are reversible, reducing anxiety and supporting experimentation without fear of permanent consequences.

3- Specific Functionality

- Key functionality within the Data Dignity application includes:
- Dashboard summaries that visualize overall data distribution and security trends
- Notification management that surfaces alerts without interrupting the user
- Data import and export tools that support transparency and data portability
- Data comparison features that allow users to evaluate data sources side by side
- Search and filtering to help users quickly locate specific data categories

4- Summary

Together, the design elements, interactions, and functionality of the Data Dignity application form a cohesive system centered on clarity and user agency. By prioritizing consistency, visibility, and low cognitive load, the interface enables users to understand and manage their personal data without feeling overwhelmed. These design decisions ensure that the application not only meets functional requirements but also delivers an experience that is approachable, empowering, and inclusive.

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